

OPF Girls College F-8/2: First Term Examination 2026

Class M-II (Science Group) · papers held August 2026 · analysis prepared 7 September 2026

The short version. Seemab sat her first term exams in four science subjects. The question papers were photographed on 6 September and I have now read all four of them. The headline is good news: every single question on all four papers sits inside the syllabus we have been drilling since April. There was nothing from an unstudied chapter, nothing from outside the confirmed boundaries, and several questions were near-identical in shape to problems from our own practice papers.

One thing to be aware of. The Mathematics paper contained a compass-and-straightedge construction question (Unit 10, Practical Geometry). We have been excluding that unit from every practice paper since July because Seemab has no geometry box. It appeared only as an optional alternative this time, and she took the other option, so it cost her nothing. But it confirms that her school does examine Unit 10, so the exclusion is now a live risk rather than a safe simplification.

What I do *not* have: the Section A objective sheets (the MCQ papers) were not photographed, and no marked scripts or results have come back yet. So this is an analysis of what was asked, not of how she did.

1. What was found and where it now lives

The papers were in the iPhone backup at `~/ImmichData/Library/iphone/2026/2026-09-06/`, five photos taken at 22:59 on 6 September (IMG_0549 to IMG_0553). I have filed them as PDFs under `seemab-reports/school/2026-08-first-term-exam/`.

Subject	Photo	Filed as	Time	Marks (B + C)
Chemistry SSC-I	IMG_0549	chemistry-paper.pdf	2 hr 15 min	27 + 12 = 39
Biology	IMG_0550	biology-paper.pdf	2 hr 15 min	27 + 12 = 39
Physics	IMG_0551, IMG_0552	physics-paper.pdf	2 hr 30 min	27 + 12 = 39
Mathematics	IMG_0553	math-paper.pdf	2 hr 15 min	24 + 16 = 40

All four papers use the same shape: Section B is short questions, each printed with an *OR* alternative, and Section C is long questions, again in *OR* pairs. Seemab's pen ticks on the papers show which alternative she picked in most rows, and there is rough working in the margins, so the papers double as a partial record of her choices.

Two gaps in the record. Section A (the objective / MCQ sheet) is a separate paper for every subject and none of the four were photographed. And the marked answer scripts have not come home, so there are no results to work from yet. If Seemab can photograph the Section A sheets, I can complete the coverage picture; the MCQ section is where she has historically been strongest and it is the part I cannot currently see.

2. Chemistry: entirely within Ch 1 to 9

Section B ran nine short questions, Section C two long ones. Mapped to her chapters:

Q	Asked	OR alternative	Chapter
1	Concentrated vs dilute solutions	Empirical formula of nicotine ($C_{10}H_{14}N_2$) and caffeine ($C_8H_{10}N_4O_2$)	Ch 2 / Ch 6
2	Colloidal dispersion vs true solution	Effect of temperature on solubility	Ch 2
3	Why is an atom electrically neutral	Define oxidation number; common oxidation state of group 1A and VIA in binary compounds	Ch 3 / Ch 7
4	Arrange in increasing ionization energy: Li, Na, K (with reason)	Arrange in decreasing shielding effect: Cl, Br, I	Ch 4
5	Why transition elements show variable oxidation state	Formation of the cation in (a) Al (b) Ca	Ch 4 / Ch 3
6	Convert to ionic equation: $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$	Balanced equation for burning hydrogen to form water	Ch 6 / Ch 7
7	Derive the formula of aluminium sulfate	Oxidation state of nitrogen in HNO_3 and NO_2	Ch 6 / Ch 7
8	Enthalpy of $H_2 + Cl_2$ from bond energies (H-H 436, Cl-Cl 243, H-Cl 432 kJ/mol)	Role of a catalyst in a chemical reaction	Ch 8
9	Forward vs reverse reactions, with example	Effect of temperature on the cobalt chloride equilibrium	Ch 9
C1	Identify oxidizing and reducing agents in three reactions	Formation of the covalent bond in N_2 , NH_3 , O_2	Ch 7 / Ch 5
C2	Exothermic vs endothermic + labelled reaction pathway diagram	Mass of 3×10^{10} molecules of oxygen gas	Ch 8 / Ch 6

Verdict: clean. Chapters 2, 3, 4, 5, 6, 7, 8 and 9 all appear; Ch 1 appears only via the objective sheet, if at all. That is exactly the Ch 1–9 range confirmed by her teacher in June and exactly what chemistry exam-12 through exam-20 have been built against. The bond-energy enthalpy calculation, the mole/Avogadro mass calculation, oxidation-state assignment and the cobalt chloride equilibrium are all topics our coverage tracker records as tested repeatedly.

Worth noting: chapters 8 and 9 are short and our tracker has been warning for weeks that their content pool is nearly exhausted, to the point where exam-17 gave Ch 9 no questions at all. The school gave both a real slot. That warning was about our ability to write *fresh* questions, not about whether the chapters matter, and this paper confirms they matter.

3. Biology: the 7.6 boundary held exactly

Q	Asked	OR alternative	Chapter
i	One significance each of palaeontology, molecular biology, immunology	Cause, vector and symptoms of dengue fever	Ch 1
ii	Structural conformation of proteins	Main features of the Watson and Crick model of DNA	Ch 6
iii	Why is it impossible to eradicate malaria	How collaboration helps scientists make new discoveries	Ch 1
iv	Structure of a nucleotide	Briefly explain the types of RNA	Ch 6
v	What makes red blood cells suitable for oxygen transport	Main types of plastids and one functional feature of each	Ch 5 / Ch 3
vi	Why respiratory and cardiovascular systems work together	Three advantages of homeostasis	Ch 5
vii	How mitosis serves as the basis of asexual reproduction	How meiosis leads to genetic variation	Ch 4
viii	Why viruses are at the borderline between living and non-living	Differentiate Domain Bacteria and Domain Eukarya	Ch 2
ix	Classify metabolic reactions A and B (figure) and tabulate the differences	Explain non-competitive inhibition of an enzyme	Ch 7 (7.1–7.3)
Q3	Effect of temperature and pH on enzyme activity	Central dogma and the role of the gene in protein synthesis	Ch 7 / Ch 6
Q4	Use of the biological method to understand the spread of malaria	Draw and explain the phases of Meiosis-I	Ch 1 / Ch 4

Verdict: clean, and the boundary call was right. Chapter 7 appears three times — catabolism vs anabolism, non-competitive inhibition, and enzyme activity vs temperature and pH — and every one of those sits in sections 7.1 to 7.6. Photosynthesis (7.7) and respiration (7.8) do not appear anywhere on the paper. The teacher's June instruction that Chapter 7 runs "till 7.6" was accurate and our decision to exclude 7.7 and 7.8 from every practice paper since then was correct.

Question ix is a figure-reading question: the paper prints two diagrams (molecules breaking into smaller units releasing energy, and smaller units building into molecules consuming energy) and asks her to classify them. Seemab has labelled them catabolism and anabolism on the paper. Our exam-content requirement of putting diagrams and figures into her biology papers is clearly the right call, since the school does it too.

4. Physics: Units 1 to 6 and Unit 9, and one near-exact repeat of our own question

Q	Asked	OR alternative	Unit
i	Scientific notation: 1168×10^{-27} , 725×10^5 , 0.02×10^{-8}	Chocolate wrapper $6.7 \text{ cm} \times 5.4 \text{ cm}$, area to the correct significant figures	U1
ii	Does a speedometer measure speed or velocity? Justify	Person weighs 600 N on Earth; weight on the Moon where $g = 1.62 \text{ m/s}^2$	U2 / U3
iii	Distance vs displacement	1200 kg car on a 60 m diameter roundabout at 25 km/h; centripetal force	U2 / U4
iv	Terminal velocity, plus the v-t graph of a paratrooper	Why a helmet is safe on a motorcycle, supported by a mathematical relation	U2 / U3
v	Liquid of three times the density of mercury: height of the liquid column in a barometer	Rolling vs sliding friction	U5 / U4
vi	Why water tanks are built at the highest level in houses	Torque on a 20 cm spanner with 50 N; spanner length to loosen the same nut with 100 N	U5 / U4
vii	Hydraulic lift, 0.002 m^2 narrow and 0.9 m^2 wide cylinder; force to lift an 1800 kg car	Convert 170 km/h to m/s; 1 m/s^2 to km/h^2 ; 36 m/s to km/h	U5 / U1–2
viii	Conditions for maximum and minimum work done	Why a girl standing in a school bus falls forward under sudden braking	U6 / U3
ix	Theory vs law, with two examples each	Plot d-t graphs: constant speed, decreasing speed, at rest	U9 / U2
Q3	Explain the fundamental forces in nature in detail	Car covers a curvy 800 m track in 40 s, straight-line distance 600 m in 36 s; find average speed and average velocity	U3 / U2
Q4	250 g at 65 cm and 100 g at 80 cm on a rod pivoted at 50 cm; where does a third 400 g mass balance it	Accuracy vs precision	U4 / U1

Verdict: clean, and one direct hit. Section B question (v) asks for the height of a barometer column for a liquid three times the density of mercury. Physics exam-14, which she sat in August, contained the same $\rho_1 h_1 = \rho_2 h_2$ barometer question with a different-density liquid. That was written specifically to close a gap the tracker had been flagging since exam-10. It landed.

Two other alignments worth recording. First, *theory vs law with examples*: our tracker marks hypothesis/theory/law as heavily saturated (tested in exams 01, 02, 07, 08 and 09) and we had been deliberately avoiding it as over-drilled. The school asked it, so five repetitions was not waste. Second, the principle of moments question in Section C is a three-mass balance, which is a harder shape than the two-mass versions in most of our papers, though exams 13 and 14 both carried the principle-of-moments statement.

Unit 6 gets only a single light slot (conditions for maximum and minimum work). Units 7 and 8 do not appear, consistent with them being outside the confirmed syllabus.

5. Mathematics: Units 1, 2, 4, 5 and 7, plus a Unit 10 construction

Q	Asked	OR alternative	Unit
i	Simplify $(81)^n \cdot 35 - (3)^{4n-1}(243)$ over $(9^{2n})(3^3)$	Simplify $\sqrt{[(216)^{2/3} \times (25)^{1/2} / (0.04)^{-3/2}]}$	U1
ii	Use the laws of logarithms to evaluate $\log_2 5\sqrt{3}$	Factorize $4x^2 - y^2 - 2y - 1$	U2 / U4
iii	HCF by factorization of $ax^2+2a^2x+a^3$, $2ax^2-4a^2x-6a^3$, $3(ax+a^2)^2$	LCM of $x^2-12xy+16y^2$, $x^2-6xy+8y^2$, $3x^2-12y^2$	U4
iv	Show A(-1,2), B(7,5), C(-2,6) are the vertices of a right-angled triangle	A(-2,1) and B(6,5) are opposite ends of a diameter; find the midpoint	U7
v	First polynomial x^2-x-6 , HCF $(x-3)$, LCM $x^3+11x^2+12x-12$; find the other polynomial	Reduce $(x^3-8)/(x^2-4) \times (x^2+6x+8)/(x^2-2x+1)$ to lowest form	U4
vi	Solve and check $\sqrt{z/(z+3)} = \sqrt{((z+2)/(z+6))}$	Construct ΔXYZ where $XY = 5$ cm, $\angle X = 30^\circ$, $\angle Z = 105^\circ$	U5 / U10
Q3	Evaluate by logarithms: $\sqrt{37.54} \times (38.4)^{1/2}$ over $\sqrt[3]{31.5}$	Simplify $(x+3)/(x^2-3x+2) + (x+2)/(x^2-4x+3) + (x+1)/(x^2-5x+6)$	U2 / U4
Q4	Cubical container of volume $(x^3-6x^2y+12xy^2-8y^3)$ m ³ : find side length, base area, total surface area, and painting cost at Rs 50/m ²	Square root by division method of $x^2/y^2 - 10x/y + 27 - 10y/x + y^2/x^2$	U4

The one real finding: Unit 10 is examinable. Section B (vi) offered a compass-and-straightedge construction as its OR alternative. Since 2 July we have excluded Unit 10 (Practical Geometry) from every math paper because Seemab has no geometry box, and the tracker records that exclusion explicitly. This paper shows the school does set it. She was not hurt this time, because it was the alternative and she took the radical equation instead, but she was effectively down to one choice on that row while her classmates had two. If a future paper puts a construction in the compulsory slot, she has no answer.

Recommended: buy a geometry box. It is a cheap fix for a risk that is now demonstrated rather than hypothetical. Once she has one, I can lift the exclusion and start building Unit 10 questions back into the practice papers.

Also worth noting: the Section C container question (find the side, base area, total surface area and painting cost of a cube from its volume as a cubic expression) is exactly the "application of factorization in daily life" material from Exercise 4.8 that exam-13 and exam-14 were built to cover. Our tracker records the cubical tank problem in exam-13 and the veranda and machine-surface problems in exam-14. That preparation was directly on target.

The mark structure differs from our practice papers: the school's Section B carries 4 marks per question over 6 questions, and Section C carries 8 marks per question over 2. Our papers have been using 3-mark short questions and 6-to-8-mark long ones. Worth matching in the next math build so the length of answer she practises matches what the school expects.

6. Summary and what to do next

Subject	Chapters examined	Anything out of our scope?
Chemistry	Ch 2, 3, 4, 5, 6, 7, 8, 9	No
Biology	Ch 1, 2, 3, 4, 5, 6, 7 (7.1-7.6 only)	No
Physics	U1, 2, 3, 4, 5, 6, 9	No
Mathematics	U1, 2, 4, 5, 7, 10	Yes: Unit 10 construction (as an OR option)

Actions

- **Buy a geometry box.** The Unit 10 exclusion is no longer safe. Once she has one, Unit 10 goes back into the math papers.
- **Get the Section A sheets photographed.** All four objective papers are missing from the record, and the MCQ section is normally her strongest. Without them the coverage picture is incomplete.
- **Get the marked scripts when they come back.** Only then can this become a results analysis rather than a paper analysis.
- **Match the school's mark structure in the next math paper** (6 × 4 in Section B, 2 × 8 in Section C) so the answer lengths she practises match what is expected.
- **Keep chapters 8 and 9 of chemistry in rotation** despite the tracker's saturation warning. The school gave both real slots.